

Changes in biotic soundscape using urbanisation and an avian invasion gradient

This project aims to work on homogenization in birds at the scale of biotic soundscape, using passive acoustic monitoring with an avian invasion and an urbanization gradient.

The goal is to develop the analysis process with one well surveyed island and then complexify the dataset using several islands.

With two main factors :



Urbanization



Avian invasion degree (AID)

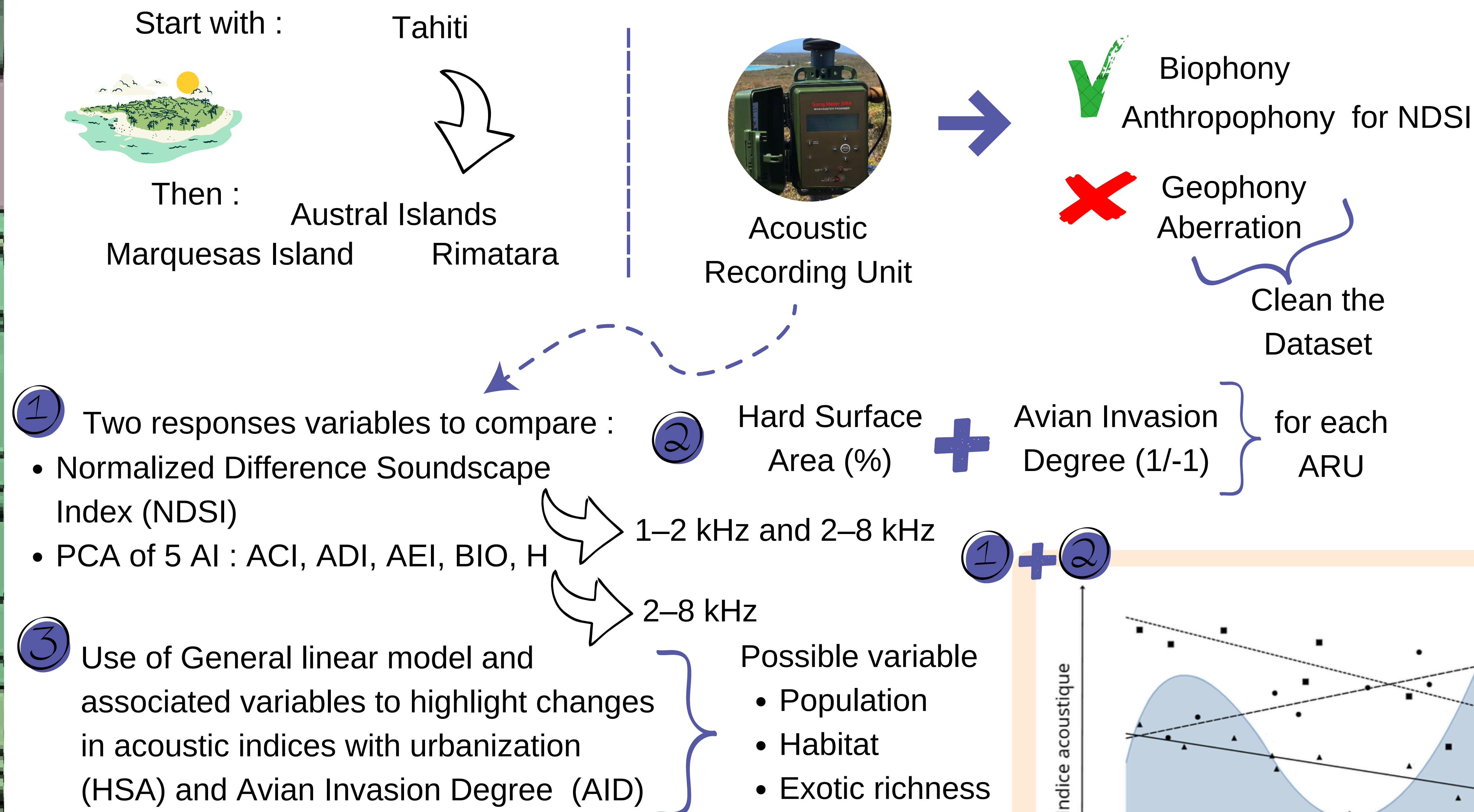
Well-marked changes in acoustic indices (AI) and bird vocalization (diversity - intensity - complexity) throughout a gradient of urbanization of the ARU.

Hypothesis

Changes on the temporality (chorus) of chorus with urbanization gradient

AI peaks follow the AID score. High in invasive, the invasive drive the AI and low in invasive, the native drive the AI

Methods and Analysis



Goal

Asses some impacts

- Several indices to compare the biotic soundscape homogenization between islands and archipelago with urbanization and avian composition.
- Used these indices to assess the difference in acoustic impact between introduced species on avian community between islands.
- Visualized change in the periodicity of soundscape between island and avian composition

Find good data
threament

- Filter the data to have reliable and precised acousitc dataset.

Perspective

- Package overlap pour regarder le chevauchement si on arrive à séparer les espèces.